PROSPECTIVE PHD RESEARCH SUMMARY

From Retrieval to Commitment

Selective and revisable claim verification under evolving evidence

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Core thesis. A reliable verifier should decide when evidence justifies commitment, maintain a provenance- and time-aware state that can be revised, and ensure later explanations remain faithful to the active state.

Problem and opportunity

Automated fact-checking is commonly evaluated as a single episode: retrieve evidence, predict a verdict, and generate a justification. Real evidence environments do not remain fixed. Evidence can be insufficient, ambiguous, copied across dependent sources, corrected, withdrawn, or rendered stale by later observations. A system may therefore produce the right answer for the wrong evidence, preserve a stale conclusion after a correction, or maintain an accurate internal record while generating an unsupported explanation.

The proposed PhD studies verification as a sequential commitment process. Its contribution is a longitudinal task and controlled evaluation connecting three capabilities that prior work usually evaluates separately. It does not claim that abstention, iterative retrieval, provenance, temporal verification, or belief revision is individually new.

Research questions

1. Selective commitment: which signals support search, deferral, commitment, or revision under explicit risk, coverage, and retrieval-cost constraints?
2. Longitudinal state: which representations remain stable under irrelevant or duplicated evidence but respond correctly to decisive corrections, contradictions, and temporal expiry?
3. State-faithful communication: when do state-derived generation constraints reduce unsupported, stale, or false-premise-following outputs without reducing usefulness?

Proposed programme

Work package	Scientific output	Principal comparison
1. Longitudinal testbed	300-500 claim trajectories with three to five timestamped snapshots; transition taxonomy; annotation and leakage audit	Static snapshots, authentic transitions, and controlled evidence interventions
2. Selective	Interpretable policies over SEARCH, DEFER,	Fixed budgets, confidence abstention,

Work package	Scientific output	Principal comparison
commitment	COMMIT, and REVISE	insufficient-evidence and iterative-retrieval baselines
3. Revisable state and outputs	Provenance-rich ledger or graph plus state-derived answer contract	Stateless recomputation, overwrite, voting, temporal graph, and truth-maintenance-inspired baselines

The core uses fixed, timestamped textual and time-series evidence so experiments remain reproducible. Live-web, multimodal, reinforcement-learning, and professional-user-study extensions are optional rather than thesis-critical.

Layered evaluation

- Commitment: risk-coverage curves, false commitment, unnecessary deferral, calibration, evidence sufficiency, and retrieval cost.
- State: transition accuracy, responsiveness, stability, temporal validity, provenance and lineage fidelity, and stale-active evidence.
- Communication: claim-evidence support, citation precision and coverage, ambiguity disclosure, retired-evidence use, false-premise resistance, and stratified human usefulness ratings.
- Inference: trajectory-level splits, bootstrap intervals, paired comparisons, cost-matched baselines, mechanism-focused ablations, and predefined failure categories.

Prospective fit with Professor Vlachos's programme

The direction is motivated by public work from Professor Vlachos's group on real-world claim verification and evidence evaluation (AVeriTeC and Ev2R), ambiguity (AmbiFC and Dis2Dis), time-series evidence (TSVer), and incremental evidence timelines (PledgeTracker). His public projects page lists automated fact-checking as a PhD topic. The proposed connecting problem is how a verifier commits, revises, and communicates across successive evidence states. This is a fit hypothesis for discussion, not evidence of supervision availability or permission to name him.

Candidate preparation and scientific distinction

My Master's work concerns governance for personalised persistent LLM memory. It developed a controlled 180-scenario diagnostic, explicit abstention and supersession decisions, provenance and lineage validation, and separate decision, state, and answer evaluation. The LLM controller achieved 88.9% decision accuracy against 42.2% for naive overwrite; independent annotation reached Cohen's kappa of 0.981. These results demonstrate task-design and evaluation preparation, not a solution to the proposed PhD.

The PhD is scientifically distinct: it studies public evidence rather than user memory, open-world acquisition rather than supplied records, source dependence and temporal validity rather than record replacement alone, and fact-checking verdicts rather than personalisation. Related work in ontology-guided

representation learning and reproducible research software supports the structured-state and engineering components.

First-year deliverables

1. Months 1-6: prior-art audit, transition taxonomy, baseline reproductions, archived evidence pipeline, and a pilot of at least 100 trajectories or 300-400 states.
2. Months 7-12: annotation validation, preregistered protocol, baseline release, layered failure analysis, and a first paper centred on the task and diagnostics.

Curated verified scholarship

Akhtar, M., Schlichtkrull, M., and Vlachos, A. (2026). Ev2R: Evaluating Evidence Retrieval in Automated Fact-Checking. TACL. https://aclanthology.org/2026.tacl-1.25/

Chen, Y., et al. (2025). PledgeTracker: A System for Monitoring the Fulfilment of Pledges. EMNLP System Demonstrations. https://aclanthology.org/2025.emnlp-demos.64/

Strong, M., and Vlachos, A. (2025). TSVer: A Benchmark for Fact Verification Against Time-Series Evidence. EMNLP. https://aclanthology.org/2025.emnlp-main.1519/

Zhu, X., Zhang, C., Chi, Y., Stafford, T., Collier, N., and Vlachos, A. (2026). Demystifying Multi-Agent Debate: The Role of Confidence and Diversity. Findings of ACL. https://aclanthology.org/2026.findings-acl.1694/

Staliunaite, I., Cheng, J., and Vlachos, A. (2026). Uncertainty Quantification for Evaluating Gender Bias in Machine Translation. Findings of EACL. https://aclanthology.org/2026.findings-eacl.116/

Bibliographic note: the two 2026 ACL entries above follow the primary ACL Anthology records, which supersede conflicting inherited audit notes. This is a curated selection, not a complete publication list.

Application status

No supervision agreement is implied. Before a formal Cambridge application names a supervisor, the applicant must discuss the research, obtain permission to name that person, confirm that the research can be undertaken in the Department, and confirm that the prospective supervisor can accept new PhD students.